

Raghav Senthil Kumar

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Built the document search engine's **REST APIs** in *Python (FastAPI)* for a competitive high school debate research SaaS platform hosted on *AWS*, developed by an *Agile* team of **10 engineers** and drawing **10K+ site visitors**
- Indexed **600K+ debate-evidence documents** on *Elasticsearch*, serving search results in **under 1 second**
- Designed an **information retrieval** algorithm that extracts evidence segments from debate cases by validating source URLs and filtering outliers with **IQR** on the word-count distribution of tag, citation and evidence text
- Developed a de-duplication algorithm that flagged **30%** of evidence documents as identical, reducing index bloat

Product Development Intern | Jetson

January 2025 – February 2025

- Redesigned user onboarding on *Figma*, cutting it from **12 to 5 pages**, validated via **A/B testing** on 30 test users
- Conducted **7 iterations of UI/UX testing** across 3 experimental features on the Jetson App (100K+ users)

Data Analytics Intern | STEM E

June 2024 – July 2024

- Analyzed 3 years of TikTok data (**200+ posts**) with *Python (pandas)*, *Hootsuite*, *Power BI* and *Microsoft Excel*
- Pitched a “fun facts” strategy to the Executive Director, who adopted it; engagement rose **24% (8K+ views)**

PROJECTS

Chronos | [Live Website](#) | [Source Code](#)

December 2025

- Built a **scheduling agent** (*React/Firebase*) that auto-books group meetings (30+ ppl) via the *Google Calendar API*
- Designed a **5-step Python scheduling algorithm** that scans calendar availability, returns a ranked list of time slots and sends email confirmation for each booking. Validated it with **14 unit and integration test cases**
- Licensed Chronos to **8 startups** & signed **2 LOIs worth \$3.5K**. **Won 2025 Congressional App Challenge**

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Stored paper data on *Supabase (PostgreSQL)* and wrote **SQL** queries to filter papers by field, year and citation count
- Built a dual-API pipeline resolving papers via *OpenAlex REST API* & backfilling metadata from *Semantic Scholar*
- Deployed a **RAG agent** (*Claude Sonnet 4.5*) that retrieves session-scoped vector documents (*Supernemory*)

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Built a *Python FastAPI* backend that ingests live water, waste & energy data from all 3 tools, runs **4 Random Forest models** to detect anomalies & deliver actionable steps via **Gemma Voice LLM** on a *Next.js* dashboard
- Led a team of 5 across project ideation, development and presentation. **Finalist at USAII Global Hackathon**

EcoBin | [Research Paper](#) | [Source Code](#)

March 2026 – May 2026

- Designed a two-stage *EfficientNetV2-S* neural network (**4.39M** parameters) in *PyTorch*, trained on **15K+ images** to classify waste into 4 disposal pathways at **96.13% accuracy**. Published a paper on *arXiv* and 2 datasets on *Kaggle*
- Engineered a smart bin prototype using *Arduino Uno* that sorts the waste item with **less than 2 second latency**

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Designed a *JSON schema* for a **Claude Sonnet 4.6 agent** to extract condition, brand, model and year from photos
- Led team of 4 to build a pricing algorithm for **Kamion (YC S22)**. **2nd of 7 at UIUC Founder's Hackathon**

SKILLS

Languages: Python, SQL, Java, C++, JavaScript, HTML/CSS

Frameworks & Databases: FastAPI, Flask, React, Next.js, Node.js, PostgreSQL, Elasticsearch, Supabase, Firebase

Cloud & Tools: REST APIs, Git, GitHub, Unit Testing, AWS, Docker, CI/CD, Google Cloud, Vercel, Agile

AI/ML: PyTorch, scikit-learn, pandas, NumPy, Hugging Face, Claude API, LLMs, RAG